Revision chemistry — Study guide

Grade 7 · IB MYP 2 Sciences · Unit 1 Chemistry's Secrets Unlocked · whole Chemistry unit

Name: _____ Class: _____ Date: _____

How to use this study guide

- It helps you revise the whole Chemistry unit before your **test**.
- Write your answers in the **My answer** column. Use full sentences and chemistry words.
- When you check your work with the answer sheet, write what you got right (✔️) or need to fix in the **Self-check** column.

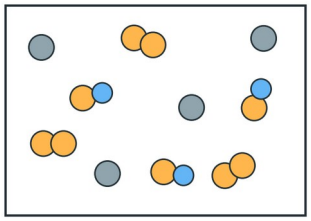
What I need to know — checklist

Tick each goal you are confident about now. Come back and tick the rest after you finish the study guide.

- ☐ I can define atom, element, compound and mixture, and give two examples of each.
- ☐ I can explain the difference between a compound and a mixture.
- ☐ I can describe the properties of acids and alkalis and name everyday examples.
- ☐ I can use the pH scale (0–14) to say if something is acidic, neutral or alkaline, and how strong it is.
- ☐ I can state the colours of universal indicator and litmus in acids and alkalis.
- ☐ I can describe neutralisation and write its word equation.
- ☐ I can name the salt made when an acid reacts with an alkali.
- ☐ I can use all of this to explain real-life situations.

A · Atoms, elements, compounds and mixtures

#	What I need to know (question or situation)	My answer	Self-check
A1	**State** what an atom is.		
A2	**Define** an element. Give two examples with their chemical symbols.		
A3	**Define** a compound. Give two examples with their formulas.		
A4	**Define** a mixture. Give two everyday examples.		
A5	**Explain** two differences between a compound and a mixture (think about fixed amounts and how easily they can be separated).		
A6	**Sort** these into element, compound or mixture: oxygen (O_2), water (H_2O), air, sodium chloride (NaCl), iron (Fe), salt water, carbon dioxide (CO_2), orange juice.		
A7	Iron is grey and magnetic. Sulfur is yellow. Iron sulfide is a black solid that is not magnetic. **Explain** what this tells you about the properties of a compound compared with the elements it is made from.		
A8	*Situation:* A gold ring is labelled "24 carat – pure gold". A cheaper ring is "9 carat": gold mixed with copper and silver. Which ring is an element and which is a mixture? **Explain** your answer.		

#	What I need to know (question or situation)	My answer	Self-check
A9	*Situation:* You spilt salt into a tray of sand and water. **Describe** the steps you would use to get back the dry salt. Name each separation method.		
A10	*Situation:* The diagram shows a box of particles: some single circles, some pairs of joined identical circles, and some joined circles of two different colours. **Explain** whether this box shows a pure substance or a mixture. 		

B · Acids and alkalis

Complete the comparison table first, then answer the questions.

Property	Acids	Alkalis
pH range		
Colour of blue litmus paper		
Colour of red litmus paper		
Taste (never test in the lab!)		
How they feel on skin		
Three everyday examples		
Three lab examples		

#	What I need to know (question or situation)	My answer	Self-check
B1	**State** what an alkali is. How is it different from a base?		

#	What I need to know (question or situation)	My answer	Self-check
B2	**Outline** why strong acids and strong alkalis both have a hazard symbol for "corrosive".		
B3	*Situation:* Your friend says, "Acids are always dangerous." Use two foods or drinks to **explain** why this is not true.		
B4	*Situation:* During a practical, a student spills dilute sodium hydroxide on their hand. **Describe** what they should do and **explain** why.		
B5	*Situation:* You have two colourless liquids, one acid and one alkali, but the labels fell off. **Describe** a safe test to find out which is which and what result you would see.		
B6	*Situation:* Old-fashioned soap made from strong alkali can make skin feel dry and sore. **Suggest** why.		

C · The pH scale

Fill in the universal indicator table.

pH	0–2	3–6	7	8–11	12–14
Universal indicator colour					
Strong / weak acid, neutral, weak / strong alkali					
One example					

#	What I need to know (question or situation)	My answer	Self-check
C1	**State** what the pH scale measures and its range.		

#	What I need to know (question or situation)	My answer	Self-check
C2	**State** the pH of pure water and what we call a substance with this pH.		
C3	Put these in order from most acidic to most alkaline: oven cleaner (pH 13), lemon juice (pH 2), pure water (pH 7), milk (pH 6.5), baking soda solution (pH 9), stomach acid (pH 1.5).		
C4	**Explain** why universal indicator is more useful than litmus paper.		
C5	*Situation:* Normal rain has a pH of about 5.6. Rain near a factory has a pH of 4. Which is more acidic? **Suggest** one effect this rain could have on statues or lakes.		
C6	*Situation:* Blueberries grow best in soil of pH 4.5–5.5. A gardener tests the soil: it is pH 7. Is the soil suitable? **Suggest** what kind of substance the gardener should add and why.		
C7	*Situation:* Your stomach contains hydrochloric acid at about pH 2. **Suggest** why your body needs such a strong acid there.		

D · Neutralisation

#	What I need to know (question or situation)	My answer	Self-check
D1	**Define** neutralisation.		
D2	Complete the general word equation: acid + alkali → _____ + _____		

#	What I need to know (question or situation)	My answer	Self-check
D3	You slowly add sodium hydroxide to hydrochloric acid with universal indicator in it. **Describe** how the colour and the pH change until the solution is neutral.		
D4	*Situation:* A bee sting is acidic. A wasp sting is alkaline. Which would you treat with baking soda (pH 9) and which with vinegar (pH 3)? **Explain** using the word *neutralisation*.		
D5	*Situation:* Indigestion is caused by too much stomach acid. **Explain** how an indigestion tablet (a base such as magnesium hydroxide) helps.		
D6	*Situation:* A farmer's field is too acidic for crops. The farmer spreads lime (calcium hydroxide) on it. **Explain** why, and **suggest** what could happen if the farmer added far too much.		
D7	*Situation:* Bacteria in your mouth make acid that damages tooth enamel. Toothpaste is slightly alkaline. **Explain** how brushing helps protect your teeth.		

E · Naming salts

How to name a salt

A salt's name has two parts: **1st name** = the metal from the alkali (or base) · **2nd name** = comes from the acid.

Practise and check your salt names in the **Salt naming simulation**: scan the QR code or go to **school.taneski.com/science/g07/01-salt-naming/**.

Scan to open

Acid	Second part of the salt name
Hydrochloric acid	
Sulfuric acid	
Nitric acid	

Complete each word equation.

#	Acid	Alkali / base	Salt	Other product
E1	hydrochloric acid	sodium hydroxide		
E2	sulfuric acid	potassium hydroxide		
E3	nitric acid	sodium hydroxide		
E4	hydrochloric acid	calcium hydroxide		
E5	sulfuric acid	magnesium hydroxide		
E6		potassium hydroxide	potassium chloride	water
E7	nitric acid		calcium nitrate	water

#	What I need to know (question or situation)	My answer	Self-check
E8	**State** the chemical name of table salt and which acid and alkali can make it.		
E9	*Situation:* A fertiliser bag says it contains potassium nitrate. **Suggest** which acid and which alkali could be used to make it.		

Unfamiliar situation challenge

You have not seen these situations before. Use everything from Sections A–E.

Situation 1 — The swimming pool. A pool manager must keep the water between pH 7.2 and 7.8. Today the test shows pH 8.4 and swimmers complain of itchy eyes.

#	Question	My answer	Self-check
S1a	Is the pool water acidic, neutral or alkaline today?		
S1b	**Suggest** whether the manager should add a dilute acid or an alkali. **Explain** why.		
S1c	The manager adds dilute hydrochloric acid, and some sodium hydroxide is in the water. Name the salt that forms.		
S1d	**Explain** why the manager should add the chemical slowly and test the pH again, rather than pour in a lot at once.		

Situation 2 — The mystery liquids. A lab technician tests four unlabelled liquids with universal indicator.

Liquid	Colour with universal indicator
P	red
Q	green
R	purple
S	yellow

#	Question	My answer	Self-check
S2a	Estimate the pH of each liquid and say if it is acidic, neutral or alkaline.		
S2b	Which liquid could be pure water? Which could be lemon juice? **Explain** your choices.		

#	Question	My answer	Self-check
S2c	The technician mixes liquids P and R in the right amounts. **Predict** the colour of the indicator at the end and **explain** what happened.		
S2d	Which liquid would be the most dangerous to spill? **Justify** your judgement.		

Add your own situation

Write three real-life situations where this chemistry is used: at home, in the kitchen, in sport, in nature or in a job. Swap with a partner: answer theirs in the *Partner's answer* column, then check each other's work.

Ideas to get you started: cleaning a kettle with vinegar, an ant bite, a cake rising with baking soda, a fish tank, a car battery.

#	My situation and question	Topic (A–E)	Partner's answer	My feedback to my partner
1				
2				
3				

Self-reflection

Question	My answer
Two things I am confident about	
Two things I still need to revise	
One situation question I found hard, and why	
My plan to revise before the test	